BMO
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Rotman

Volatility Trading Case

Rotman International Trading Competition 2026

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OVERVIEW

The Volatility Trading Case gives participants the opportunity to generate profits by implementing options strategies to trade volatility. The underlying asset of the options is a non-dividend-paying Exchange Traded Fund (ETF) called RTM that tracks a major stock index. Participants will be able to trade shares of the ETF and 1-month call/put options at 10 different strike prices. Information including the ETF price, options prices, and news releases will be provided. Participants are encouraged to use the provided information to identify mispricing opportunities and construct options trading strategies accordingly.

DESCRIPTION

There will be 2 heats and teams will allocate 2 team members for each heat. Each participant may compete in only one of the 2 heats. Each heat will consist of 8 independent sub-heats, with each sub-heat representing one month of calendar time.

Parameter	Value
Number of trading sub-heats	8
Trading time per sub-heat	300 seconds (5 minutes)
Calendar time per sub-heat	1 month (20 trading days)

News will be released during each sub-heat. Order submissions using the RIT API will be enabled. Data retrieval via Real-time Data (RTD) Links and the RIT API will also be enabled.

MARKET DYNAMICS

Participants will be able to trade RTM and 20 separate options contracts on RTM at the beginning of the case. All options are European, so early exercise is not allowed.

Call Ticker	Strike Price	Put Ticker
RTM1C45	45	RTM1P45
RTM1C46	46	RTM1P46
RTM1C47	47	RTM1P47
RTM1C48	48	RTM1P48
RTM1C49	49	RTM1P49
RTM1C50	50	RTM1P50
RTM1C51	51	RTM1P51
RTM1C52	52	RTM1P52
RTM1C53	53	RTM1P53
RTM1C54	54	RTM1P54

All securities are priced by market-makers who will always quote a bid-ask spread of 2 cents (i.e. \$49.99×\$50.01 for the RTM). The bids and asks are for very large quantities (there are no liquidity constraints in this case).

The price of RTM follows a random-walk and the path is generated using the following process:

$$P_{\mathrm{RTM,t}} = P_{\mathrm{RTM,t-1}} \times (1 + r_{\mathrm{t}}), \text{where } r_{\mathrm{t}} \sim N(0, \sigma_{\mathrm{t}})$$

The price of the stock is based on the previous price multiplied by a return that is drawn from a normal distribution with a mean of zero and standard deviation (volatility) on an annualized basis.

Your analysts have developed a very effective model to forecast the realized volatility of the underlying security RTM; they will be sending you weekly updates. The market maker quotes prices for options using the Black-Scholes pricing model and a variance risk premium equal to zero. Therefore, in equilibrium the implied volatility associated with an options price will be equal to the realized volatility of the underlying security.

However, it has been observed in this market that the volatility forecasts made by the market maker are often less informed than those provided by the informed analysts. Therefore, after a discrete change in the realized volatility of the underlying, the market maker's volatility forecast embodied in the quoted options price will not accurately reflect the volatility of the underlying security (RTM). Nevertheless, as time goes on, the market maker learns about the underlying volatility – at least until a new shock to the realized volatility occurs.

Your task is to look for mispricing opportunities in the options markets and profit from that mispricing. When mispricing occurs, you should take a position in the options and at the same time trade the underlying security to hedge your delta exposure in order to build a delta neutral portfolio.

For example, if a call option has an implied volatility that is 5% greater than the analysts' forecasted realized volatility of the ETF, the option is considered overpriced. In this case, you should write call options and then hedge the delta exposure by trading the underlying asset appropriately.

For ease of computation, one trading year is assumed to be 240 days ($20 \text{ days } \times 12 \text{ months}$). The case represents one month (20 days) of calendar time. The trading period is divided into 4 weeks, with $t = 1 \ldots 75$ being week one, $t = 76 \ldots 150$ being week two, and so on. At the beginning of each week, the volatility of RTM (σ_t) will shift and the new value will be provided to participants. In addition, at the middle of each week (e.g. $t = 37$) an analyst estimate of next week's volatility value will be announced.

Sample News Release Schedule

Time	Week	Release
1	Week 1	The analysts have informed you that the realized volatility of RTM for this week will be 18%
1	Week 1	The delta limit for this sub-heat is 10,000 and the penalty percentage is 1%
37	Week 1	The analysts have informed you that the realized volatility of RTM next week will be between 28% and 33%
75	Week 2	The analysts have informed you that the realized volatility of RTM for this week will be 29%
…	…	…
225	Week 4	The analysts have informed you that the realized volatility of RTM for this week will be 26%

The focus of this case is on trading volatility without being exposed to price changes of the underlying security, RTM. Participants are therefore required to manage their portfolio's delta exposure. Recognizing the transaction costs and impracticality of perfect delta hedging (i.e. keeping the portfolio's delta at zero at all times), the RITC scoring committee will allow the portfolio's delta to be different from zero but it is required to stay between $-delta\ limit$ and $+\ delta\ limit$. Please note that $delta\ limit$ is an integer number greater than 1,000 that will be announced at the beginning of a sub-heat via a news release in RIT. For example, the following news could be released: "The delta limit for this sub-heat is 5,000 and the penalty percentage is 0.5%". According to that news, any participant that has a portfolio delta greater than 5,000 will be penalized at the penalty percentage of 0.5% according to the penalties explained below.

For every second that a participant exceeds the limit ($\pm delta\ limit$), participants will be charged a penalty according to the following formula:

$$Penalty\ for\ second\ t = (|Portfolio's\ Delta\ at\ time\ t| - Delta\ Limit) \times x$$

where, $delta\ limit$ is the limit imposed for each sub-heat and x is the fine rate set for each sub-heat, which will be released at the beginning of each sub-heat. Penalties will be calculated and incorporated directly into your real-time P&L in RIT, so your displayed performance reflects both trading outcomes and any applicable penalties throughout the case.

TRADING LIMITS AND TRANSACTIONS COSTS

Each participant will be subject to gross and net trading limits specific to the security type as specified below. The gross trading limit reflects the sum of the absolute values of the long and short positions across all securities in each security type; the net trading limit reflects the sum of long and short positions such that short positions negate any long positions. Trading limits will be enforced and participants will not be able to exceed them.

Security Type	Gross Limit	Net Limit
RTM ETF	50,000 Shares	50,000 Shares
RTM Options	2,500 Contracts	1,000 Contracts

The maximum trade size will be 10,000 shares for RTM and 100 contracts for RTM options. Transaction fees will be set at $0.01 per share traded for RTM and $1.00 per contract traded for RTM options. As with standard options markets, each contract represents 100 shares (purchasing 1 option contract for $0.35/option will actually cost $35 plus a $2 commission, and will settle based on the exercise value of 100 shares).

POSITION CLOSE-OUT

Any outstanding position in RTM will be closed at the end of trading based on the last-traded price. There are no liquidity constraints for the options nor RTM. All options will be cash-settled based on the following upon expiration:

$$\text{Call Option Payout} = \max\{0, S - K\},$$
$$\text{Put Option Payout} = \max\{0, K - S\},$$

where,

S is the last price of RTM,
K is the strike price of the option.

KEY OBJECTIVES

Objective 1

Build a model to forecast the future volatility of the underlying ETF based on known information and given forecast ranges. Participants should use this model with an options pricing model to determine whether the market prices for options are overvalued or undervalued. They should then trade the specific options accordingly.

Objective 2

Use Greeks to calculate the portfolio exposure and hedge the position to reduce the risk of the portfolio while profiting from volatility differentials across options.